# STM32F469I-DISCO

## MB1189

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Sheet 8: USB_OTG_FS

Sheet 9: Audio Codec and Microphone MEMS

Sheet 10: Power

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Sheet 13: Extension Connectors


## Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

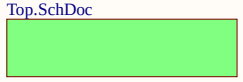
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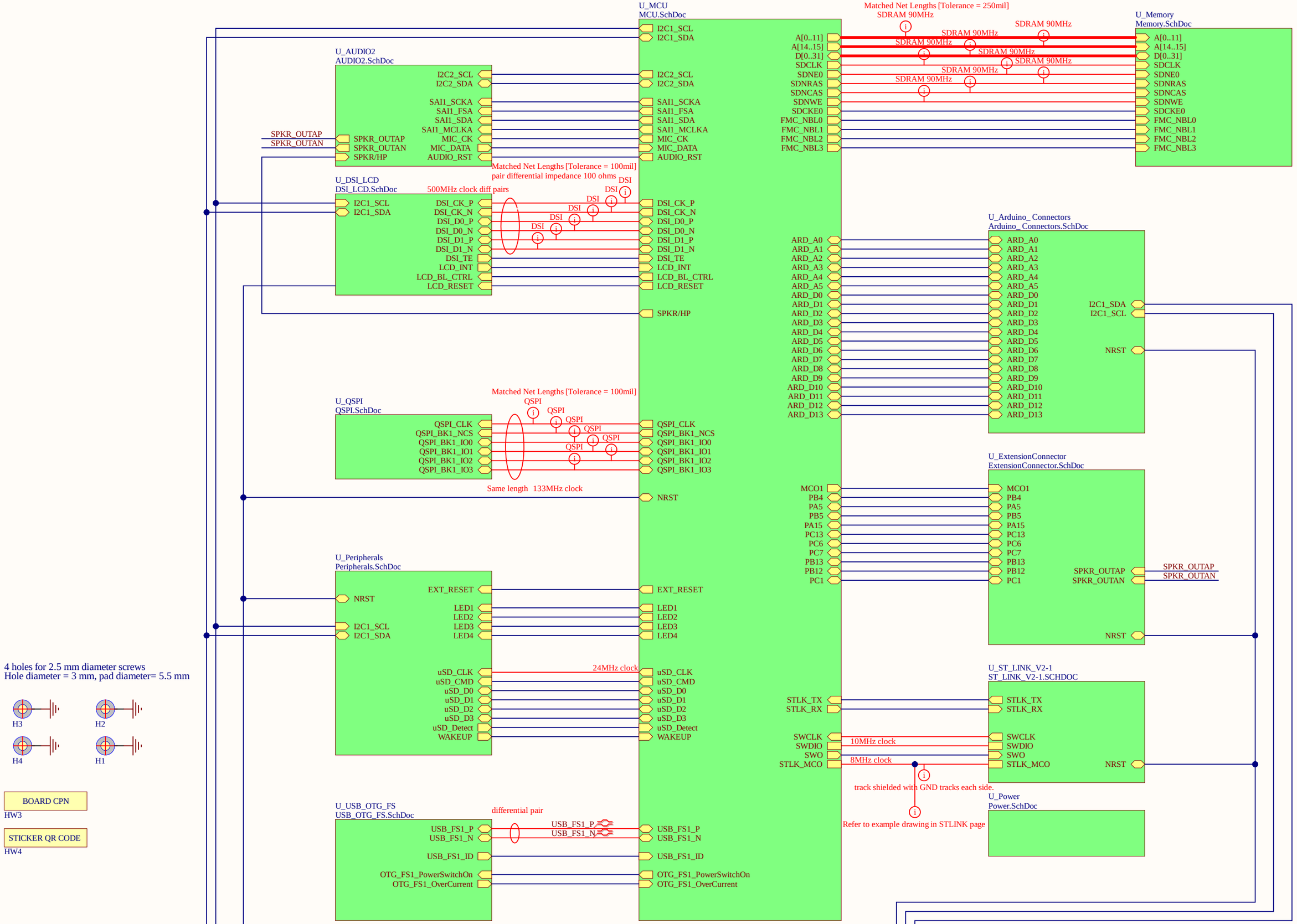
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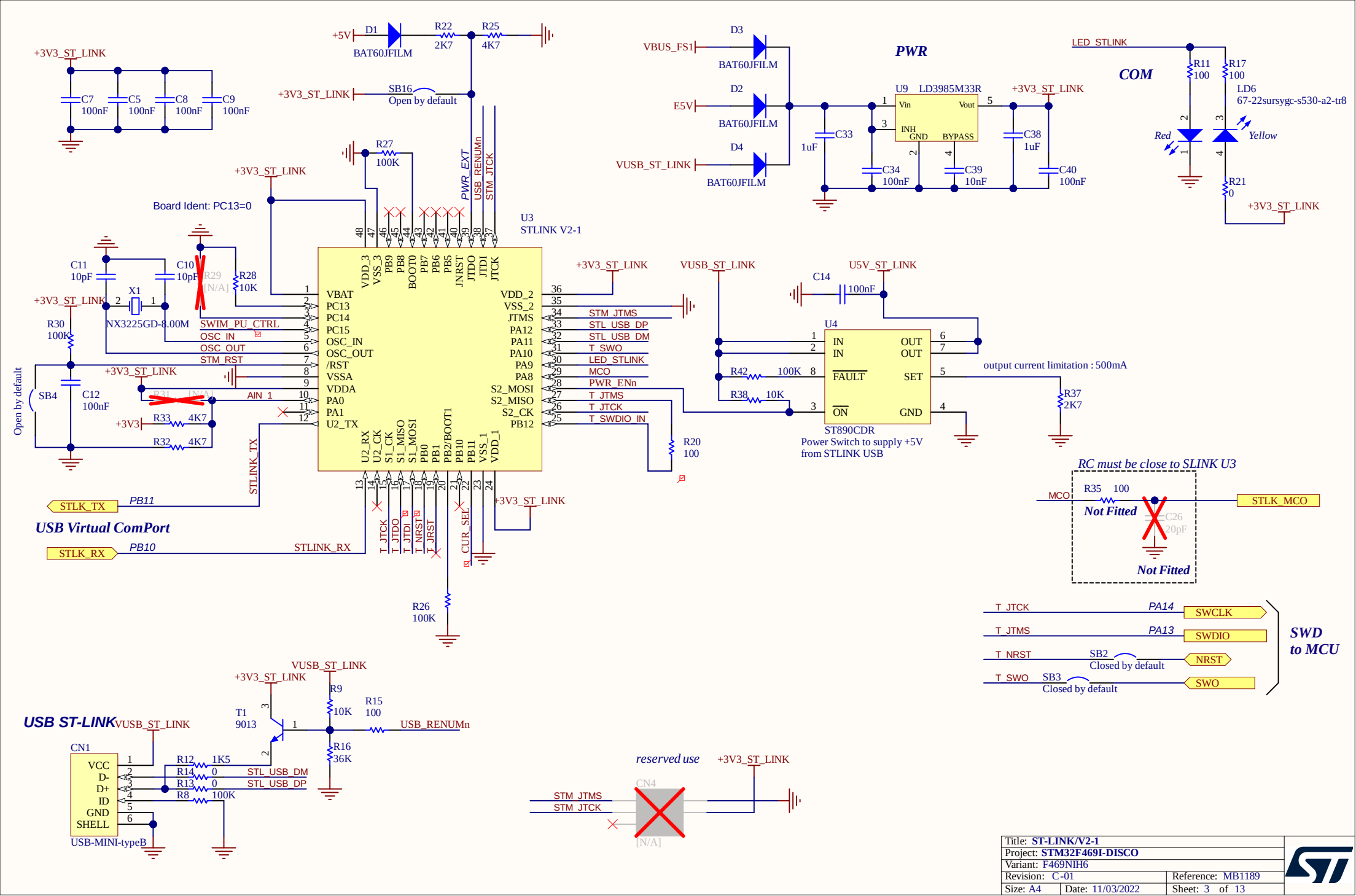

U_Top
Top.SchDoc



| | | | |
|---|---|---|---|
| Title: **Project overview** | | | |
| Project: **STM32F469I-DISCO** | | | |
| Variant: F469NIH6 | | | |
| Revision: C-01 | | Reference: MB1189 | |
| Size: A4 | Date: 11/03/2022 | Sheet: 1 of 13 | |



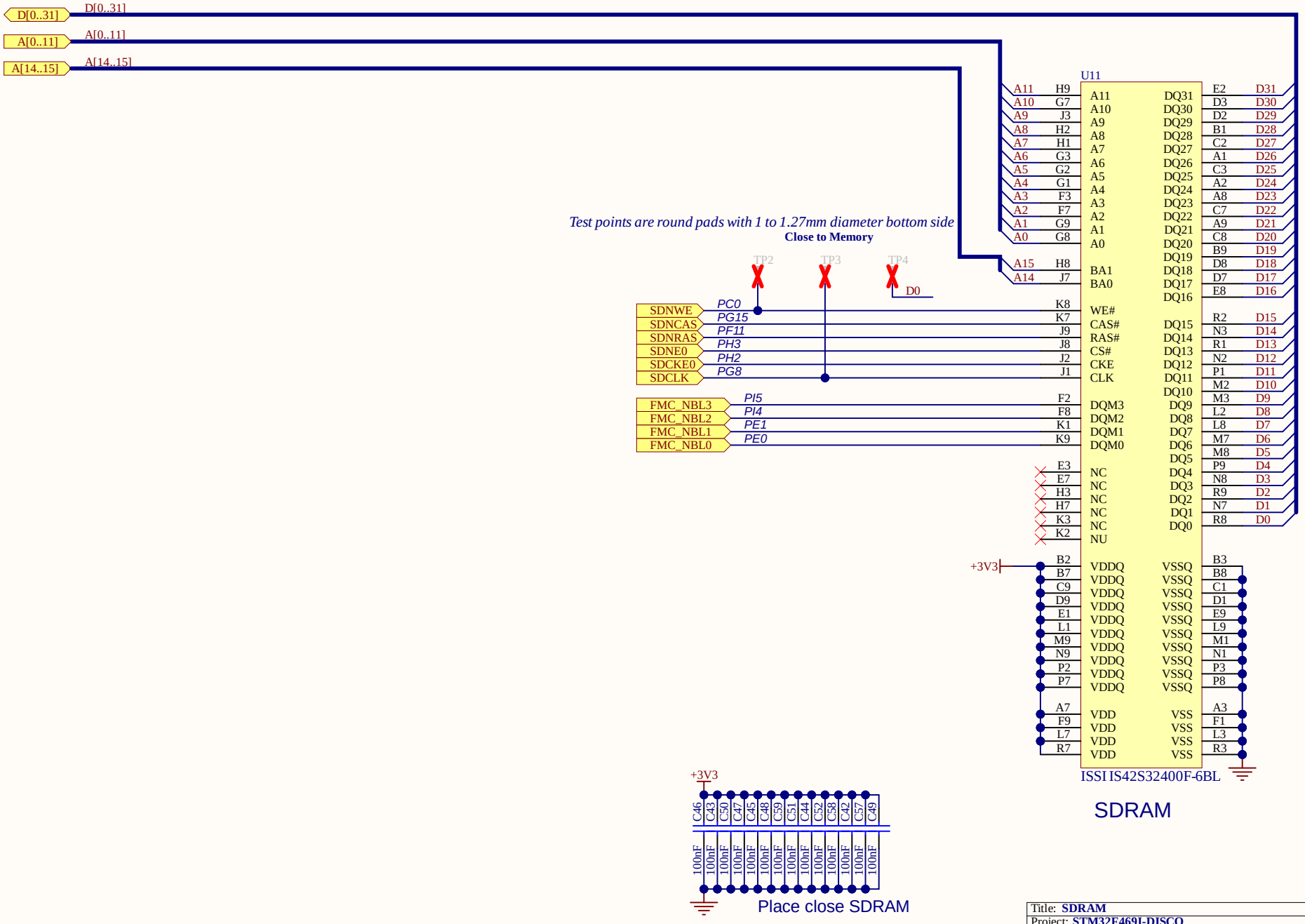

| | | |
|---|---|---|
| Title: **MB1189** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A3 | Date: 11/03/2022 | Sheet: 2 of 13 |




| | | |
|---|---|---|
| Title: **ST-LINK/V2-1** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 3 of 13 |

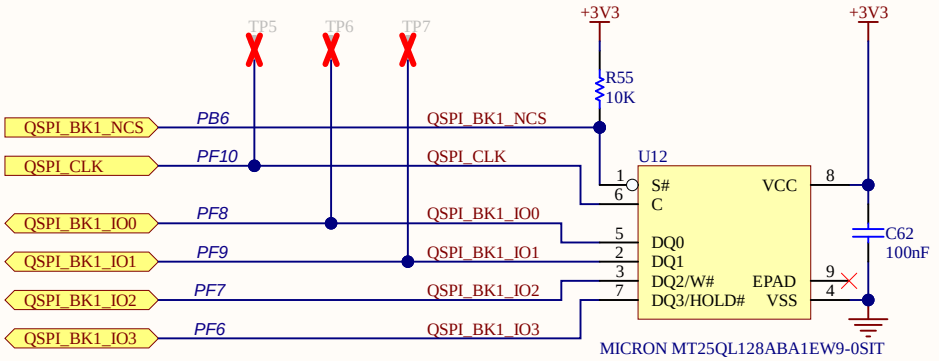
WAKEUP
ARD_D3
ARD_D5
ARD_A5
ARD_D6
ARD_D9
MCO1
LCD_BL_CTRL
PA5
USB_FS1_ID
USB_FS1_N
USB_FS1_P
VBUS_FS1
SWDIO
SWCLK
PA15
EXT_RESET
ARD_A0
OTG_FS1_PowerSwitchOn
I2S3_CK
SB6
Closed by default
SWO
SB7
Open by default
PB4
PB5
QSPI_BK1_NCS
OTG_FS1_OverCurrent
I2C1_SCL
I2C1_SDA
+3V3
R1401K5
R142 1K5
STLK_RX
STLK_TX
PB12
PB13
ARD_D12
ARD_D11
C93 3.0pF
X3
NX3215SA-32.768K
C94 3.0pF
C85 8.2pF
X2
NX3225GD-8.00M
C83 8.2pF close to MCU
R132 0
R133 0
R131 0
NRST
BOOT0
R143 510
R150 [N/A]
SB15
Closed by default
R149 10K
R144 [N/A]
STLK_MCO
SB19
Open by default
PH0
SDNWE
ARD_A1
ARD_A2
ARD_A3
ARD_A4
PC6
PC7
PC13
LED4
DSI_TE
LCD_INT
DSI_CK_P
DSI_CK_N
DSI_D0_P
DSI_D0_N
DSI_D1_P
DSI_D1_N
to EXT Connector
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8 PC9 PC10 PC11 PC12 PC13
PH0 PH1
PJ0 PJ1 PJ2 PJ3 PJ4 PJ5 PJ12 PJ13 PJ14 PJ15
PK3 PK4 PK5 PK6 PK7
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
PA0-WKUP
PDR_ON
STM32F469NIH6
U14A
R135 0
uSD_D0
uSD_D1
uSD_D2
uSD_D3
uSD_CLK
PA11
PA12
PC8
PC9
PC10
PC11
PC12
PJ0
PJ1
PJ2
PJ3
PJ4
PJ5
PJ12
PJ13
PJ14
PJ15
PK3
PK4
PK5
PK6
PK7
R87 and R135 are for reserved use
U14B
uSD_CMD
ARD_D13
LED2
LED3
MIC_DATA
MIC_CK
R87 0
TP1
I2S3_CK
Close to MCU
FMC_NBL0
FMC_NBL1
AUDIO_RST
SPKR/HP
SAI1_FSA
SAI1_SCKA
SAI1_SDA
QSPI_BK1_IO3
QSPI_BK1_IO2
QSPI_BK1_IO0
QSPI_BK1_IO1
QSPI_CLK
SDNRAS
uSD_Detect
LED1
SAI1_MCLKA
SDCLK
ARD_D0
ARD_D8
ARD_D7
ARD_D4
ARD_D2
ARD_D1
ARDUINO USART6_RX
ARDUINO USART6_TX
SDNCAS
SDNE0
SDCKE0
R121 2K2
R124 2K2
+3V3
I2C2_SCL
I2C2_SDA
I2C2 to codec: <100kHz
ARD_D10
LCD_RESET
FMC_NBL2
FMC_NBL3
PD0 PD1 PD2 PD3 PD4 PD5 PD6 PD7 PD8 PD9 PD10 PD11 PD12 PD13 PD14 PD15
PE0 PE1 PE2 PE3 PE4 PE5 PE6 PE7 PE8 PE9 PE10 PE11 PE12 PE13 PE14 PE15
PF0 PF1 PF2 PF3 PF4 PF5 PF6 PF7 PF8 PF9 PF10 PF11 PF12 PF13 PF14 PF15
PG0 PG1 PG2 PG3 PG4 PG5 PG6 PG7 PG8 PG9 PG10 PG11 PG12 PG13 PG14 PG15
PH2 PH3 PH4 PH5 PH6 PH7 PH8 PH9 PH10 PH11 PH12 PH13 PH14 PH15
PI0 PI1 PI2 PI3 PI4 PI5 PI6 PI7 PI8 PI9 PI10 PI11 PI12 PI13 PI14 PI15
PI8-ANTI_TAMP2
STM32F469NIH6
U14C
VDD_MCU
L5 BEAD(FCM1608KF-601T03)
VDDA
R123 0
VREF+
DSI_VDD12
SB14
Closed by default
C81 2.2uF
VDD1 VDD2 VDD3 VDD4 VDD5 VDD6 VDD7 VDD8 VDD_USB33 VDD10 VDD11 VDD12 VDD13 VDD14 VDD15 VDD17 VDD18 VDD19
VDDA
VREF+
VDD_VREGDSI
DSI_VDD12
DSI_VCAP
DSI_VSS
VSS1 VSS2 VSS3 VSS4 VSS5 VSS6 VSS7 VSS8 VSS9 VSS10 VSS11 VSS12 VSS13_VSS18 VSS14 VSS15 VSS17 VSS19 VSS20
VSSA
VREF-
VBAT
BYPASS-REG
VCAP1
VCAP2
R122 10K
C96 2.2uF
C76 2.2uF
Ceramic capacitor (Low ESR)
D[0..31]
A[0..11]
A[14..15]
C67 C72 C88 C90 C54 C84 C53 C71 C75 100nF
C60 4.7uF
C98 4.7uF
C65 1uF
C77 100nF
C74 C69 C95 C68 C91 C73 C89 C92 C70 100nF
C80 1uF
C78 100nF
C86 1uF
C87 100nF

| | | |
|---|---|---|
| Title: **MCU** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A3 | Date: 11/03/2022 | Sheet: 4 of 13 |


D[0..31]
A[0..11]
A[14..15]
Test points are round pads with 1 to 1.27mm diameter bottom side
Close to Memory
TP2
TP3
TP4
D0
U11
SDNWE
SDNCAS
SDNRAS
SDNE0
SDCKE0
SDCLK
PC0
PG15
PF11
PH3
PH2
PG8
FMC_NBL3
FMC_NBL2
FMC_NBL1
FMC_NBL0
PI5
PI4
PE1
PE0
+3V3
ISSI IS42S32400F-6BL
SDRAM
C46 C43 C50 C47 C45 C48 C59 C51 C44 C52 C58 C42 C57 C49
100nF
Place close SDRAM
Title: SDRAM
Project: STM32F469I-DISCO
Variant: F469NIH6
Revision: C-01
Reference: MB1189
Size: A4
Date: 11/03/2022
Sheet: 5 of 13

*Test points are round pads with 1 to 1.27mm diameter bottom side*

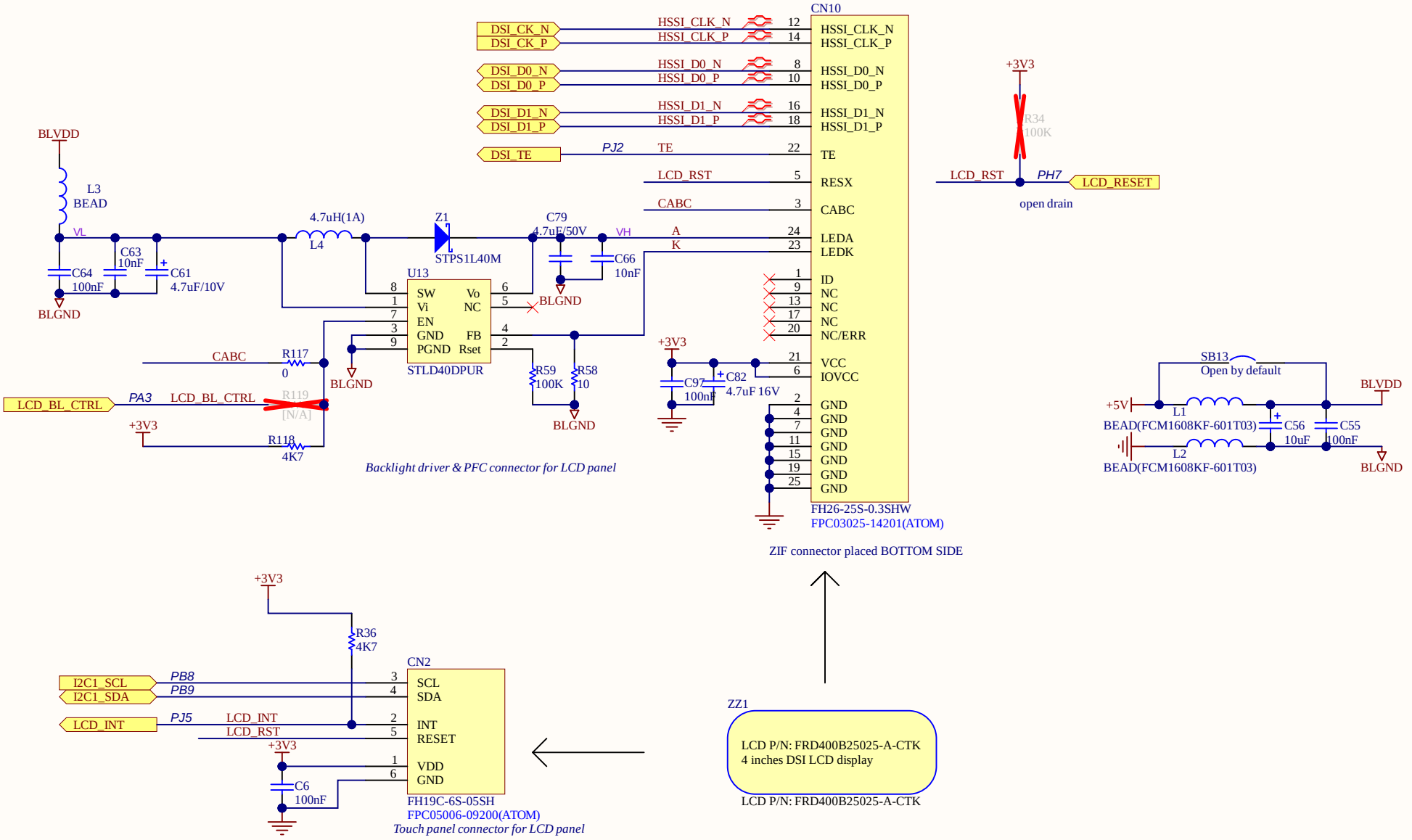
**Close to QSPI memory**




MICRON MT25QL128ABA1EW9-0SIT

Exposed central pad should not be connected to any voltage on PCB

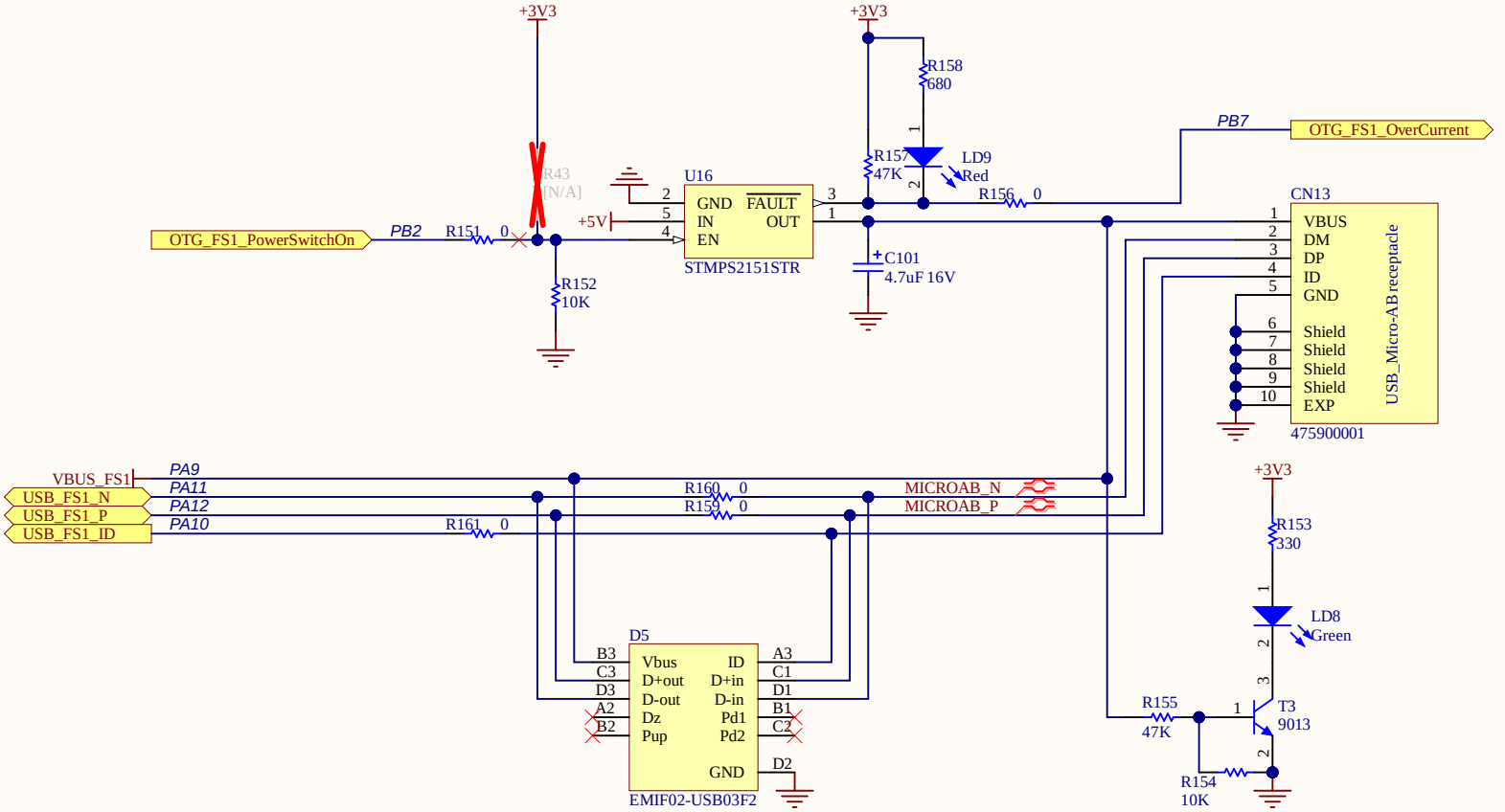
| | | |
|---|---|---|
| Title: **QSPI** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 6 of 13 |




I2C Address:0x54

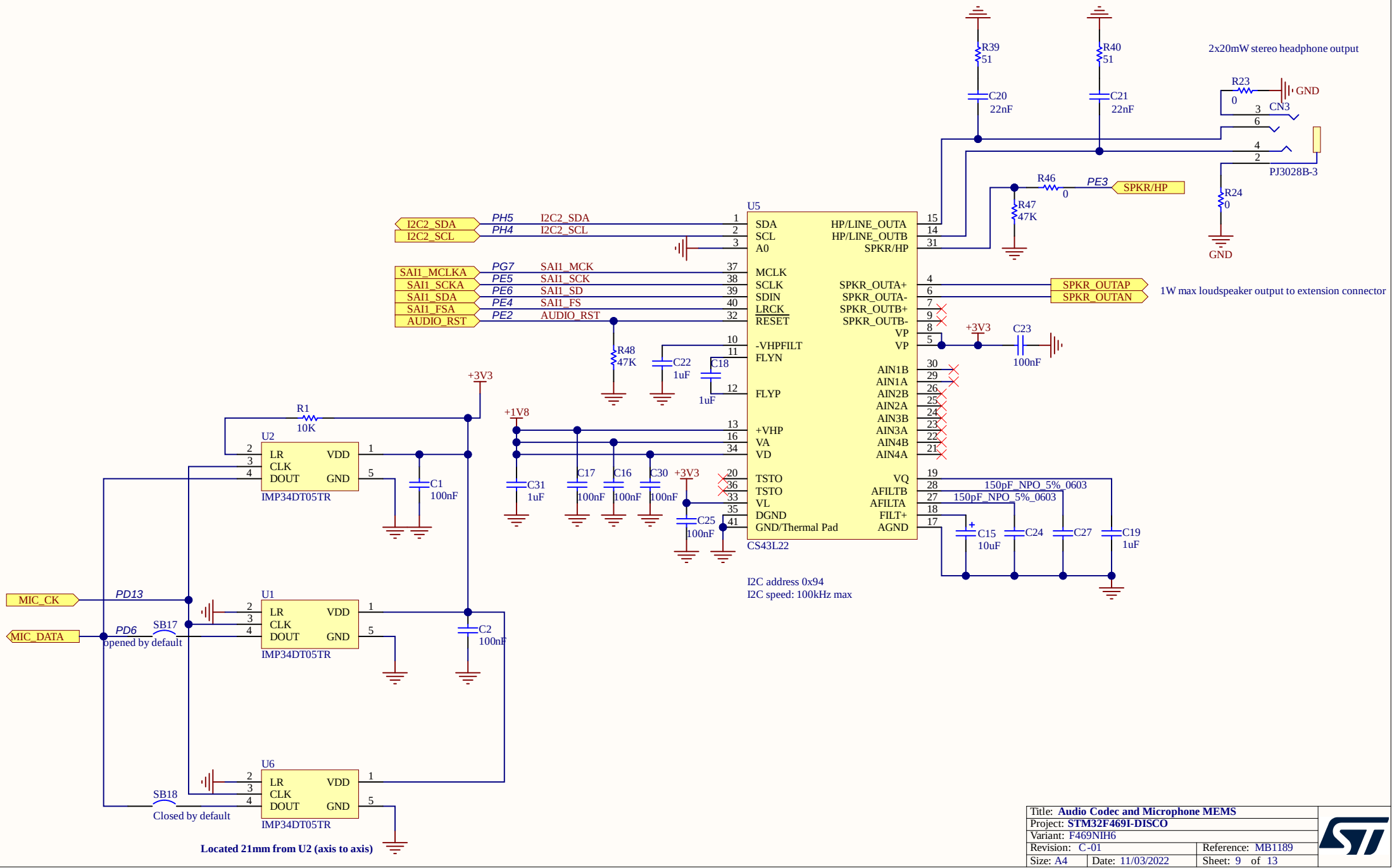
**Mechanical: 2 Slots in PCB for the 2 flex from LCD**

| | | | |
|---|---|---|---|
| Title: **4 inch DSI LCD board** | | | |
| Project: **STM32F469I-DISCO** | | | |
| Variant: F469NIH6 | | | |
| Revision: C-01 | | Reference: MB1189 | |
| Size: A4 | Date: 11/03/2022 | Sheet: 7 of 13 | |

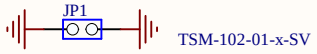


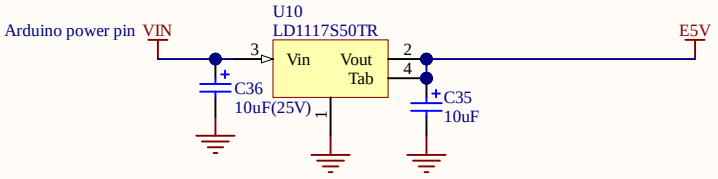

**USB Full Speed operating range voltage: 3.0V<VDDUSB<3.6V**

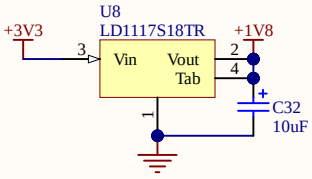
| | | |
|---|---|---|
| Title: **USB_OTG_FS** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 8 of 13 |

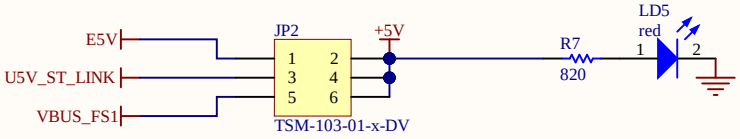


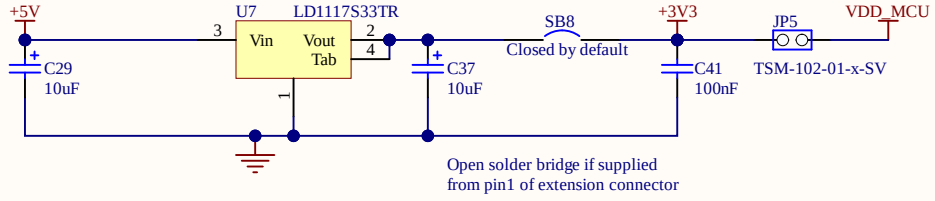

| | | | |
|---|---|---|---|
| Title: **Audio Codec and Microphone MEMS** | | | |
| Project: **STM32F469I-DISCO** | | | |
| Variant: F469NIH6 | | | |
| Revision: C-01 | | Reference: MB1189 | |
| Size: A4 | Date: 11/03/2022 | Sheet: 9 of 13 | |



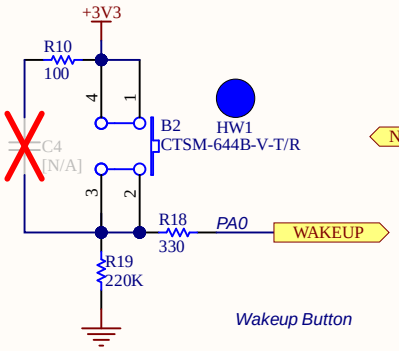



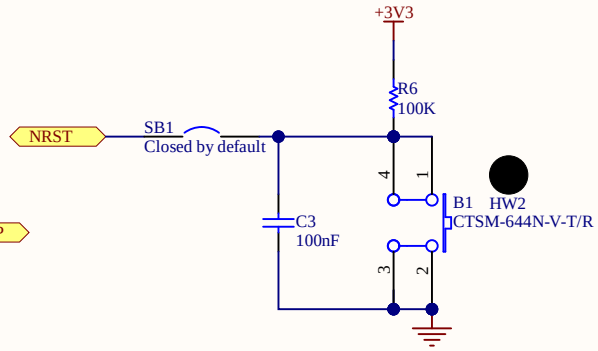

| | | |
|---|---|---|
| Title: **Power** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 10 of 13 |

## Button

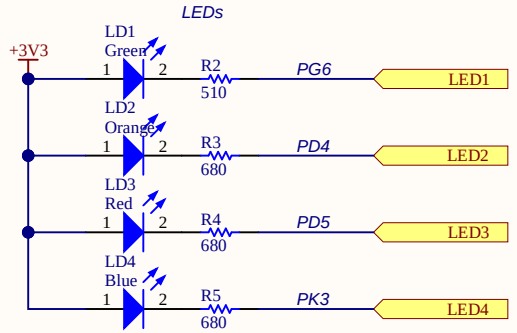



*Wakeup Button*

## Reset Button

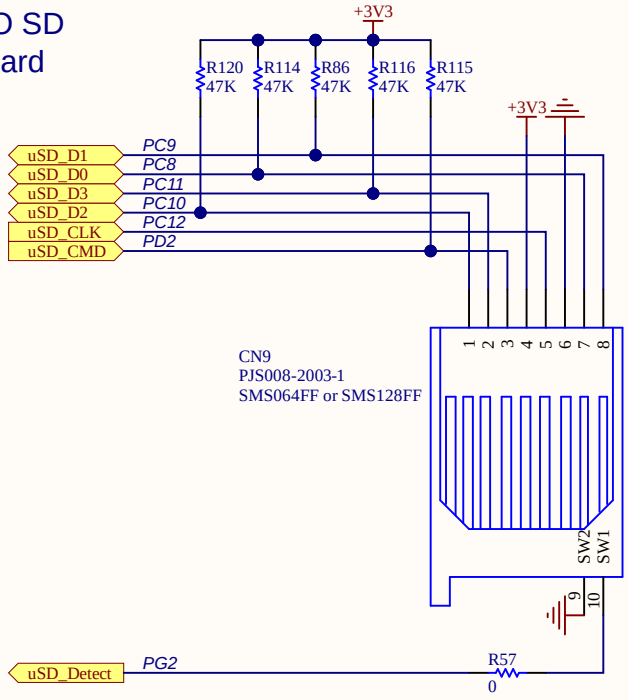



## LEDs

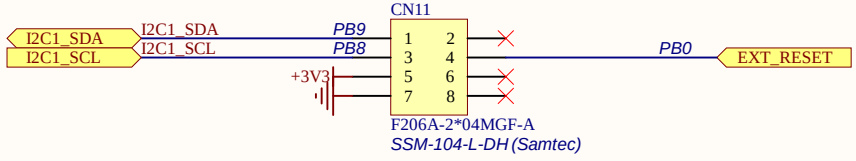



**The 4 LEDs are top side**

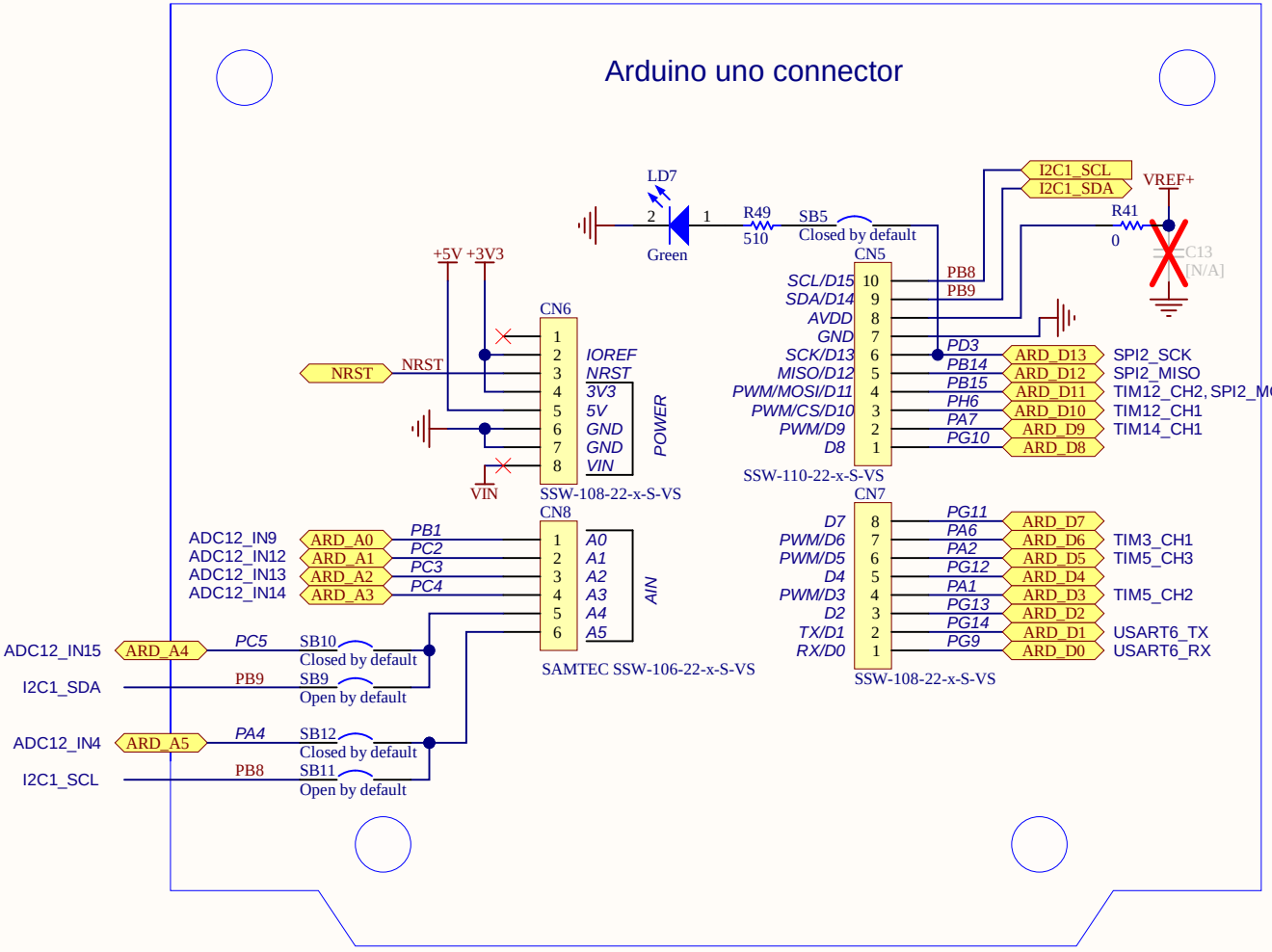
## MICRO SD (TF) Card



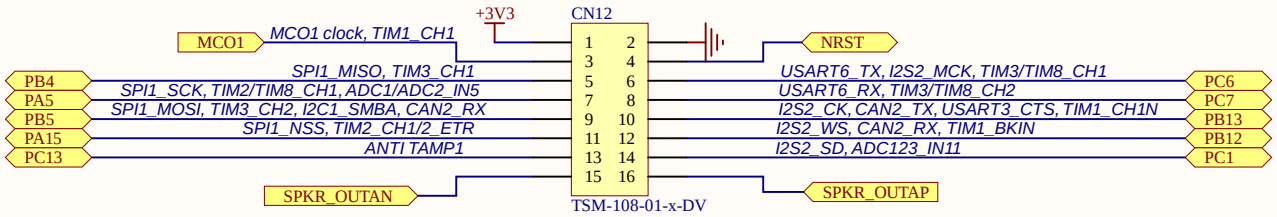

## EXT/RF E2P Connector




| | | |
|---|---|---|
| Title: **Peripherals** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 11 of 13 |



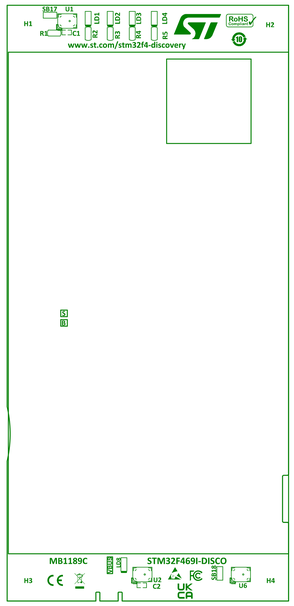

| | | |
|---|---|---|
| Title: **Arduino Uno connector** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 12 of 13 |




post heigth: 5.84mm

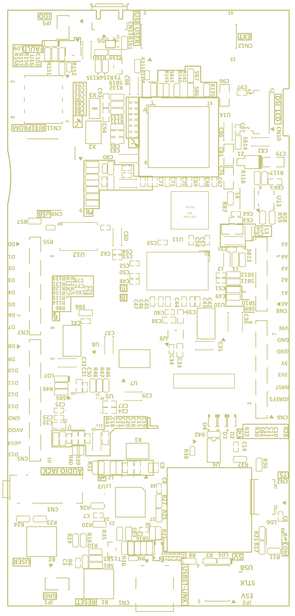
## Extension connector interfaces

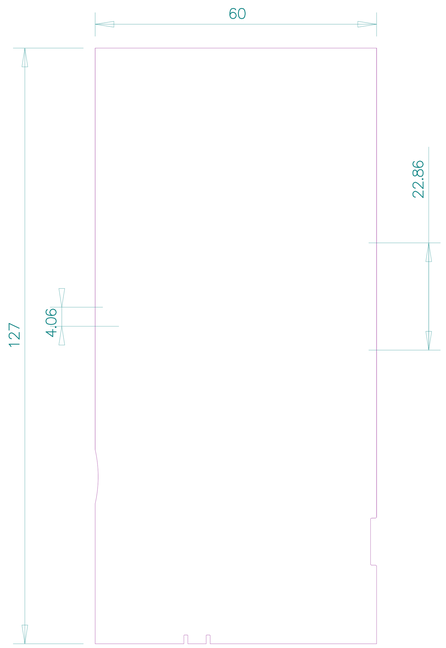
| | | |
|---|---|---|
| Title: **Extension Connectors** | | |
| Project: **STM32F469I-DISCO** | | |
| Variant: F469NIH6 | | |
| Revision: C-01 | | Reference: MB1189 |
| Size: A4 | Date: 11/03/2022 | Sheet: 13 of 13 |




| | | |
|---|---|---|
| Project: STM32F469I-DISCO | | |
| Layer: TOP SILKSCREEN | Gerber: .GTO | |
| Variant: F469NIH6 | Ref: MB1189 | |
| Date: 11/03/2022 | Rev: C | |

| | |
|---|---|
| Project: STM32F469I-DISCO | |
| Layer: BOTTOM SILKSCREEN | Gerber: .GBO |
| Variant: F469NIH6 | Ref: MB1189 |
| Date: 11/03/2022 | Rev: C |


CN11
CN12
CN7
CN3
CN1
CN2
CN4
CN5
CN6
CN8
CN9
CN10
AUDIO JACK
USB
STLK
ESV
BOOT0
RESET
JOY_SEL
JOY_DOWN
JOY_LEFT
JOY_RIGHT
JOY_UP
TAMPER
WAKEUP
OSC_IN
OSC_OUT
OSC32_IN
OSC32_OUT
VBAT
VDD
VSS
3V3
5V
GND
NRST
BOOT
SWCLK
SWDIO
SWO
MCO
MCO1
MCO2
USART_TX
USART_RX
I2C_SCL
I2C_SDA
SPI_SCK
SPI_MISO
SPI_MOSI
SPI_NSS
CAN_TX
CAN_RX
USB_DM
USB_DP
USB_ID
USB_VBUS
USB_OTG_FS_OVRCR
USB_OTG_FS_PWR_EN
USB_OTG_HS_OVRCR
USB_OTG_HS_PWR_EN
ETH_MDC
ETH_MDIO
ETH_RMII_CRS_DV
ETH_RMII_REF_CLK
ETH_RMII_RXD0
ETH_RMII_RXD1
ETH_RMII_TXD0
ETH_RMII_TXD1
ETH_RMII_TX_EN
ETH_RST
ETH_MII_COL
ETH_MII_CRS
ETH_MII_RX_CLK
ETH_MII_RXD2
ETH_MII_RXD3
ETH_MII_TXD2
ETH_MII_TXD3
ETH_MII_TX_CLK
ETH_MII_RX_DV
ETH_MII_RX_ER
ETH_MII_TX_EN
ETH_MII_RXD0
ETH_MII_RXD1
ETH_MII_TXD0
ETH_MII_TXD1
ETH_MII_PPS_OUT
ETH_MII_MDC
ETH_MII_MDIO
ETH_MII_INTN
ETH_MII_RST
ETH_MII_CRS_DV
ETH_MII_REF_CLK
ETH_MII_RXD0
ETH_MII_RXD1
ETH_MII_TXD0
ETH_MII_TXD1
ETH_MII_TX_EN
ETH_MII_RX_ER
ETH_MII_RX_DV
ETH_MII_RX_CLK
ETH_MII_TX_CLK
ETH_MII_RXD2
ETH_MII_RXD3
ETH_MII_TXD2
ETH_MII_TXD3
ETH_MII_COL
ETH_MII_CRS
ETH_MII_MDC
ETH_MII_MDIO
ETH_MII_INTN
ETH_MII_RST
ETH_MII_PPS_OUT




| | | |
|---|---|---|
| Project: STM32F469I-DISCO | | |
| Layer:FAB DARWING & DRILL LEGEND | Gerber: | |
| Variant: [No Variations] | Ref: MB1189 | |
| Date: 11/03/2022 | Rev: C | |